THE UNIVERSAL MODULE OF DIFFERENTIALS OF SHEAVES OF RINGS IS THE SHEAFIFICATION OF THE PRESHEAF OF MODULES OF DIFFERENTIALS OF THE RINGS OF SECTIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The construction of the universal module of differentials as the cokernel of an explicit map of free modules. The defining equation is reproduced verbatim as a context span of this card (source lines 4999-5011). Supplies the map whose presheaf analogue is written down in the proof.
- The construction of the module of differentials of a ring map as the cokernel of the analogous map of free modules over a ring. Invoked inside the counted proof at source lines 5070-5072 as a cross-chapter pointer. Identifies the value of the cokernel of the presheaf map with the module of differentials of the corresponding rings.

Source reference. Original reference algebra-definition-differentials: sources/stacks-modules/pinned-original-source.tex, source lines 5072–5072. Navigation pointer only: the counted text refers to the Stacks item algebra-definition-differentials, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. Let $\varphi : \mathcal{O}_1 \to \mathcal{O}_2$ be a homomorphism of sheaves of rings on X. Then $\Omega_{\mathcal{O}_2/\mathcal{O}_1}$ is the sheaf associated to the presheaf $U \mapsto \Omega_{\mathcal{O}_2(U)/\mathcal{O}_1(U)}$.*

Date: stacks-project:modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

$$\begin{array}{ccc} \mathcal{O}_2[\mathcal{O}_2 \times \mathcal{O}_2] \oplus \mathcal{O}_2[\mathcal{O}_2 \times \mathcal{O}_2] \oplus \mathcal{O}_2[\mathcal{O}_1] & \longrightarrow & \mathcal{O}_2[\mathcal{O}_2] \\ [(a, b)] \oplus [(f, g)] \oplus [h] & \longmapsto & [a + b] - [a] - [b] + \\ & & [fg] - g[f] - f[g] + \\ & & [\varphi(h)] \end{array} \tag{0.1.1}$$

Proof. Consider the map (0.1.1). There is a similar map of presheaves whose value on the open U is

$$\mathcal{O}_2(U)[\mathcal{O}_2(U) \times \mathcal{O}_2(U)] \oplus \mathcal{O}_2(U)[\mathcal{O}_2(U) \times \mathcal{O}_2(U)] \oplus \mathcal{O}_2(U)[\mathcal{O}_1(U)] \longrightarrow \mathcal{O}_2(U)[\mathcal{O}_2(U)]$$

The cokernel of this map has value $\Omega_{\mathcal{O}_2(U)/\mathcal{O}_1(U)}$ over U by the construction of the module of differentials in Algebra, Definition algebra-definition-differentials. On the other hand, the sheaves in (0.1.1) are the sheafifications of the presheaves above. Thus the result follows as sheafification is exact. □